

ANNA ROMA

ROBOTICS, CONTROL & EMBEDDED SYSTEMS
MATHEMATICAL & SOFTWARE ENGINEER

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ABOUT ME

Master's student in Software Engineering with a strong background in mathematical engineering and hands-on experience in robotics and embedded systems. In the university student Team DIANA, I work on control, state estimation and sensor fusion for autonomous systems, developing and testing real-time algorithms in MATLAB/Simulink, C and embedded development. Active member of BIS Italia (British Interplanetary Society), I am motivated by learning, experimentation and complex technical challenges.

WORK EXPERIENCE

November 2022 - Present

Team DIANA - Torino, Italy

Lead Control Systems Engineer

- Leading a team of engineering students developing mobility and robotic arm control algorithms for planetary rovers.
- Designed and implemented real-time sensor fusion pipelines (GPS, IMU, wheel odometry, stereo vision) in Simulink and C++.
- Developed a Kalman Filter achieving centimeter-level localization accuracy during autonomous navigation.
- Collaborated with electronics and firmware teams on embedded integration, data logging and sensor calibration.
- Managed the laboratory: equipment procurement, testing setup, and operations for competitions and ground-station tests.

June 2024 - Present

Bricks 4 Kids - Torino, Italy

Robotics Teacher

- Taught robotics and programming to elementary and middle school students using LEGO Spike, increasing engagement through project-based learning.

October 2024 - July 2025

Immaginazione e Lavoro - Torino, Italy

Computer Science and Robotics Teacher

- Taught programming languages, 3D printing and web development to middle school students, guiding them through the creation of functional websites.
- Delivered hands-on lessons in applied robotics, including Arduino board design, wiring and implementation of basic embedded logic.

September 2021 - September 2024

Colosseo Theater - Torino, Italy

Front House Manager

- Managed front-of-house operations during live performances, coordinating staff and ensuring smooth events in a fast-paced environment.

EDUCATION

October 2024 – Present

POLITECNICO DI TORINO

Master's Degree in Software Engineering

Field of Study: Automation and Control for Cyber-Physical Systems

September 2020 – October 2024

POLITECNICO DI TORINO

Bachelor's Degree in Mathematics for Engineering

Field of Study: Applied Mathematics, Artificial Intelligence and Modeling

Thesis: Implementation of a Kalman Filter for Position Estimation of the Ardito Rover.

PROJECTS

- Monopod Reinforcement Learning Locomotion (2024)
- Pac-Man on ARM board (2024).
- Distributed Control for CPSs (2024).
- InfoCovid-19 Web Platform (2020).

More is available on my GitHub repository.

PUBLICATIONS

A. Roma, "The Kalman Filter as a modular tool: a versatile solution for autonomous systems and beyond."

Congress: 76th International Astronautical Congress (IAC), Sydney, Australia, IAC-25-B2,IP,18,x10002.

Publisher: International Astronautical Federation (IAF).

SKILLS

- MATLAB, Simulink (advanced)
- C, C++, Python, Rust, Git (advanced)
- ARM microcontrollers, Embedded Development (proficient)
- Linux, ROS (proficient)

LANGUAGES

- Italian (native)
- English (B2)